

AIGR Deep Research Report — *S. pombe*

Rrg8 (O14106 / SPAC31G5.06)

Seed hypothesis: *Schizosaccharomyces pombe* Rrg8 is a **conserved mitochondrial RNase P-associated accessory factor required for mitochondrial tRNA 5'-end processing**.

Source review file: `genes/SCHP0/rrg8/rrg8-ai-review.yaml` (focus: free_text)

Executive Judgment

Verdict: Partially supported (orthology-based; unverified in *S. pombe*).

The hypothesis is an **accurate paraphrase of the experimentally characterized budding-yeast ortholog**, *S. cerevisiae* **RRG8 = MTA1 = YPR116W** (SGD:S000006320). In *S. cerevisiae*, that protein is (i) required for efficient 5' processing of mitochondrial tRNAs (IMP), (ii) localized to the matrix side of the inner mitochondrial membrane (IDA), and (iii) physically associated with the mitochondrial RNase P protein subunit Rpm2p (Mta1p co-immunopurifies with Rpm2p) — all from a single primary study,

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(Guedes-Monteiro et al., 2019).

For *S. pombe* **rrg8** specifically, however, **every functional and localization annotation is orthology-inferred (evidence code ISO, GO_REF:0000024) transferred from that budding-yeast gene, with no direct *S. pombe* experimental evidence** for localization, tRNA-processing function, or RNase P association.

Key caveats that a curator must weigh: - **"Conserved" is fungal-restricted, not broad.** PomBase taxonomic distribution = *fungi only*; the primary paper describes the family as *Saccharomycetaceae*; the Pfam family PF17068 ("Required for respiratory growth protein 8, mitochondrial") is detected in the budding-yeast protein but **not** matched on *S. pombe* O14106. This factor exists only in fungi that retain an **RNA-based** mitochondrial RNase P (rnpB-encoded), and has no counterpart in the protein-only PRORP systems of metazoa/plants. - **Orthology is weak and not independently corroborated.** *S. pombe* vs *S. cerevisiae* **RRG8** global identity $\approx$ 23% (Needleman-Wunsch, 65/280 aligned positions). PomBase curates them

as 1:1 orthologs (manual). But three independent checks fail to reinforce this: (i) O14106 carries **no InterPro/Pfam match** (InterPro HTTP 204), whereas Q06109 clearly has IPR031415/PF17068 ("RRG8, mitochondrial"); (ii) **AlphaFold** models diverge sharply — Q06109 is confidently folded (mean pLDDT **75.5**, 69% of residues >70) while O14106 is largely **low-confidence/disordered** (mean pLDDT **50.2**, 0% >70, 43% <50); (iii) the primary paper calls these genes "of the **Saccharomycetaceae** family," yet PomBase extends the ortholog into *S. pombe* (Taphrinomycotina). None of this **refutes** orthology (low pLDDT and an absent Pfam match can both stem from shallow MSA / a narrow family model built on Saccharomycetaceae), but it means the entire ISO transfer rests on the manual ortholog call alone and should be flagged as a **transfer-risk caveat**, not treated as settled. - **"Accessory factor" is the correct framing.** The budding-yeast molecular function is explicitly **unknown** (GO:0003674, ND); the protein is *required for* the process and *associated with* RNase P, but is **not** a catalytic RNase P subunit. Do **not** assign a ribonuclease molecular-function term. - **A phenotype discrepancy exists.** *S. pombe* rrg8Δ is annotated **inviable** in two genome-wide screens

(
P 20473289;
P 23697806),

whereas the budding-yeast null is viable but respiratory-deficient. This is a species difference (or screen artifact) and does not itself contradict the molecular role. - **A localization conflict exists.** UniProt's automated subcellular prediction for O14106 is cytoplasm/nucleus, conflicting with the ISO mitochondrial annotation. The automated prediction is low-confidence and should be overridden by the orthology-based mitochondrial call, but note that no *direct S. pombe* mitochondrial localization has been shown.

Bottom line for curation: The existing GO annotations (GO:0097745 BP; GO:0005739 CC) are **defensible ISO leads that should be retained**, with the source correctly attributed to a strong budding-yeast study. The gene-product name is appropriate. But the review should (a) qualify "conserved" as fungal-restricted, (b) flag that no direct *S. pombe* evidence exists, (c) keep molecular function unassigned, and (d) not upgrade "RNase P-associated" into a physical-complex CC term for *S. pombe* without direct data.

Evidence Matrix

Citation (PMID/DB)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
P 30759361 (Guedes-Monteiro 2019)	Mutant phenotype (IMP) + localization (IDA) + interaction (co-IP)	Supports (for ortholog)	Rrg8/Mta1 required for mito tRNA 5' processing; RNase P-associated	MTA1/MTA2/GEP5/PET130 needed for efficient 5' processing of <i>S. cerevisiae</i> mito tRNAs; inner-membrane; HMW complexes; Mta1p co-IPs with Rpm2p	<i>S. cerevisiae</i> mitochondria
SGD:S000006320 (RRG8/YPR116W/ MTA1)	Database (curated)	Supports/ Qualifies	Identity & function of ISO source	"Protein of unknown function; required for efficient 5' processing of mitochondrial tRNAs, for respiratory growth and mitochondrial genome maintenance; localizes to the matrix side of the inner mitochondrial membrane"; MF = ND	<i>S. cerevisiae</i>
PomBase SPAC31G5.06 annotation_details	Database (provenance)	Qualifies	Basis of <i>S. pombe</i> annotations	GO:0097745 & GO:0005739 both ISO / GO_REF:0000024 / with:SGD:S000006320 ; taxonomic distribution "fungi only"; ortholog = YPR116W	<i>S. pombe</i>
	Structural/ evolutionary		<i>S. pombe</i> mito RNase	Identified/confirmed mtDNA-encoded rnnpB	Fission yeast mtDNA

Citation (PMID/DB)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
P 12923256 (Seif 2003)		Supports (pathway context)	P is RNA- based	(RNase P RNA) in <i>S. pombe</i> and <i>S. octosporus</i>	
P 22991464 (Howard 2012)	Structural/ evolutionary/ review	Qualifies	RNA- vs protein-only RNase P	Yeast mito genomes encode an RNase P RNA; PRORP is a distinct, metazoan/ plant innovation	Cross-species
P 24184848 (Herbert 2013, review)	Review	Qualifies	Yeast mito RNA factors context	<i>S. pombe</i> encodes ~10 PPR proteins; mito general factors include RNA polymerase and RNase P	Yeast mitochondrion
P 20473289 ; P 23697806	HT mutant phenotype (microscopy)	Conflicts/ Qualifies	Essentiality	<i>S. pombe</i> rrg8Δ scored inviable	Genome- wide <i>S. pombe</i> deletion screens
UniProt O14106	Database (automated)	Conflicts	Localization	Automated SL prediction: cytoplasm/ nucleus; protein existence "Predicted"; no InterPro/Pfam xref	<i>S. pombe</i>

Citation (PMID/DB)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
Computed (this run)	Computational	Qualifies	Orthology strength (sequence)	Global identity $\approx 23\%$ (65/280); Q06109 has IPR031415/PF17068, O14106 has no InterPro/Pfam match (HTTP 204)	Pairwise Spombe/S.
Computed (this run; AlphaFold DB v6)	Computational (structure)	Qualifies / competing	Structural conservation	Q06109 confidently folded (mean pLDDT 75.5; 69% >70); O14106 low-confidence/ disordered (mean pLDDT 50.2; 0% >70; 43% <50)	AlphaFold monomer models

GO Curation Implications (leads requiring curator verification)

GO term	Aspect	Current evidence	Recommended action (lead)
GO:0097745 mitochondrial tRNA 5'-end processing	BP	ISO from SGD:S000006320 (donor IMP, P 30759361)	Retain as an ISO lead. It is an appropriately specific BP term backed by a strong ortholog IMP. Keep ISO evidence and the GO_REF; do not upgrade to experimental.
GO:0005739 mitochondrion	CC	ISO from SGD:S000006320 (donor IDA GO:0099617)	Retain ; optionally consider the more specific CC by orthology (donor is IDA for GO:0099617 "matrix side of mitochondrial inner membrane" / GO:0005743 inner membrane). Given ~23% identity, a conservative curator may keep the general GO:0005739.
GO:0003674 molecular function	MF	Donor = ND (unknown)	Leave unassigned. Do not add ribonuclease/nuclease activity or "protein binding." No MF is supported.
(candidate) part_of RNase P / RNase P complex association	CC/ complex	Only budding-yeast co-IP (Mta1-Rpm2)	Do not add for <i>S. pombe</i>. Physical association is shown only in <i>S. cerevisiae</i> and only for Mta1; treat as non-core, hypothesis-level.

Gene-product name "mitochondrial tRNA 5'-end processing protein Rrg8" is consistent with the evidence and can stand.

Module / holoenzyme context (Iteration 3 checks)

- The budding-yeast tRNA 5'-processing module comprises four accessory factors, each mapping to a **distinct recognized orthogroup**: MTA1/RRG8 (Q06109; OrthoDB 4035333at2759, eggNOG ENOG502S46Y), MTA2 (Q12167; 4065996at2759, ENOG502RYGE), GEP5 (Q12393; 4066262at2759, ENOG502S2Q8), PET130 (P47065; 4062049at2759, ENOG502S17D). **Limitation:** the OrthoDB v12 public API returned `null` for these v11-style group IDs in this environment, so I could **not** programmatically enumerate member species (i.e., whether *S. pombe* also retains orthologs of MTA2/GEP5/PET130). This is reported rather than inferred. Whether the *whole* module is conserved in *S. pombe* — which would

strengthen the pathway-context plausibility of *rrg8*'s role — remains an open, checkable question (see Discriminating Tests).

- A targeted PubMed search for a recent **solved structure of the yeast mitochondrial RNase P holoenzyme** naming Mta1/Rrg8 returned nothing indexed. Thus the "RNase P subunit/associated" status of Mta1/Rrg8 currently rests on the **co-IP + co-sedimentation** evidence in

P 30759361

("may be present in a common complex with Rpm2p"), not on a structural assignment. This keeps "accessory factor" (rather than defined structural subunit) the appropriately conservative wording.

Mechanistic Scope

- **Immediate molecular activity being claimed:** an accessory (non-catalytic) protein that promotes 5' endonucleolytic maturation of mitochondrial pre-tRNAs by the RNA-based mitochondrial RNase P holoenzyme (RNA subunit = mtDNA *rnpB*; protein subunit = Rpm2 in budding yeast).
 - **Direct gene-product activity (ortholog-supported):** membrane-associated component that is *required for* efficient 5' processing and *physically associates* with Rpm2p. Its own catalytic activity is unknown (ND).
 - **Downstream / indirect effects (must not be conflated with core function):** respiratory-growth deficiency, mitochondrial genome maintenance defects, decreased plasma-membrane electron transport (budding-yeast null), and the *S. pombe* inviable phenotype. These are loss-of-function consequences, not the molecular function.
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Conflicts and Alternatives

1. **Localization conflict:** UniProt automated prediction (cytoplasm/nucleus) vs orthology-based mitochondrion. Resolve in favor of mitochondrion, but acknowledge no direct *S. pombe* localization.

2. **Orthology / conservation risk:** ~23% identity, no PF17068 match on O14106, and an AlphaFold model that is low-confidence/disordered (mean pLDDT 50.2, 0% >70) versus a well-folded budding-yeast ortholog (75.5). The budding-yeast function may not transfer cleanly; the *S. pombe* protein could have diverged or acquired additional roles (consistent with its inviable phenotype). Caveat: low pLDDT/absent Pfam can also arise from shallow MSA for a fungi-restricted protein, so this weakens — but does not refute — the transfer.
 3. **Phenotypic divergence:** *rrg8Δ* inviable in *S. pombe* vs viable/respiratory-deficient in *S. cerevisiae*. Alternative explanations: (a) *S. pombe* dependence on mitochondrial translation/mtDNA maintenance makes the defect lethal; (b) an additional essential (non-mitochondrial) function; (c) genome-wide-screen false positive.
 4. **"RNase P-associated" strength:** In the primary paper, direct co-IP with Rpm2 was demonstrated for **Mta1p** (=RRG8) specifically — good for this gene — but the authors phrase the complex membership cautiously ("**may** be present in a common complex"). An alternative model is that these factors influence mito tRNA stability or membrane-tethered RNA metabolism rather than acting as bona fide RNase P subunits.
 5. **No paralog confusion detected:** single-copy in *S. pombe*; the ISO donor is unambiguously YPR116W/RRG8/MTA1.
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Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
No direct <i>S. pombe</i> mito tRNA-processing assay	PomBase annotation_details (ISO only)	The core BP claim is untested in <i>S. pombe</i>	Northern blot / RNA-seq of mito pre-tRNA 5' ends in rrg8-depletion (conditional allele, given inviability)
No direct <i>S. pombe</i> localization	UniProt (predicted cyto/nucleus); PomBase (ISO mito)	CC term unverified in the organism	GFP-tagging + submitochondrial fractionation
Does <i>S. pombe</i> have an Rpm2 ortholog / how is its mito RNase P organized?	Confirmed mtDNA rnpB exists (P 12923256); Rpm2 protein partner not established	"RNase P-associated" presupposes a protein partner to associate with	Identify <i>S. pombe</i> mito RNase P protein subunits; co-IP Rrg8 with rnpB RNA / Rpm2 ortholog
Orthology confidence (~23% id, no Pfam match, disordered AF model)	Computed NW identity; InterPro API (204/no match); AlphaFold pLDDT (O14106 50.2 vs Q06109 75.5)	Determines whether ISO transfer is safe	Foldseek/TM-align structural superposition with MSA-depth control; phylogenetic ortholog analysis across Taphrinomycotina; cross-species complementation
Basis of inviable phenotype	Two genome-wide screens	Affects interpretation of essentiality vs respiratory role	Tetrad dissection / conditional allele; test viability on non-fermentable vs fermentable media

Discriminating Tests (most efficient first)

1. **Cross-species complementation:** does *S. pombe* rrg8 rescue the respiratory defect of *S. cerevisiae* mta1Δ (and vice versa)? Directly tests functional orthology behind the ISO transfer.

2. **Mito pre-tRNA 5'-end assay in *S. pombe*** (conditional *rrg8* allele): accumulation of 5'-extended mito pre-tRNAs would confirm the transferred BP.
3. **Submitochondrial localization** of tagged Rrg8 in *S. pombe* (fractionation/protease protection) to confirm inner-membrane, matrix-side placement.
4. **Interaction test:** co-purify Rrg8 with *S. pombe* mitochondrial RNase P (*rnpB* RNA and any protein partner) to test the "RNase P-associated" claim in the organism.
5. **AlphaFold structural comparison** of O14106 vs Q06109 to buttress orthology despite low sequence identity. (*Attempted this run: O14106's model is low-confidence/disordered — mean pLDDT 50.2 — so it did not corroborate a shared fold; a Foldseek/TM-align comparison with MSA-depth controls would be more decisive.*)
6. **Module-conservation check:** determine whether *S. pombe* retains orthologs of MTA2/GEP5/PET130 (OrthoDB/eggNOG member-species enumeration, or PomBase reverse ortholog lookup). Presence of the whole module would strengthen the pathway context; absence would argue the *rrg8* role may not transfer.

Curation Leads (require curator verification)

- **Candidate reference to attach / verify:**

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(Guedes-Monteiro et al., 2019, *RNA Biol*/... — "5' processing of *Saccharomyces cerevisiae* mitochondrial tRNAs requires expression of multiple genes"). Exact snippets to verify: "Here, we identify four novel genes *MTA1*, *MTA2*, *GEP5* and *PET130* of the *Saccharomycetaceae* family that are necessary for an efficient processing of mitochondrial tRNAs." and "This is supported by pull-down assays showing co-immunopurification of *Rpm2* with *Mta1p*." (Note SGD alias: RRG8/YPR116W = MTA1.)

- **Pathway-context references:**

P 12923256

(*S. pombe* mtDNA-encoded *rnpB* / RNA-based mito RNase P);

P 22991464

(RNA-based yeast vs protein-only PRORP).

- **GO actions:** Retain GO:0097745 (BP, ISO) and GO:0005739 (CC, ISO); optionally consider the more specific CC (GO:0005743 / GO:0099617) by orthology. Keep MF unassigned. Do not add a physical RNase P-complex CC term for *S. pombe* without direct data. Avoid "protein binding" as a recommendation.
- **Qualifier for the review text:** change "conserved" → "conserved within fungi (Ascomycota; fungi-only distribution)"; add explicit note that the *S. pombe* annotation is orthology-inferred (ISO) from budding-yeast Mta1/Rrg8 and is **not** experimentally verified in fission yeast; flag ~23% ortholog identity and absent PF17068 match as a transfer caveat.
- **Suggested curator questions:** (1) Is the PomBase 1:1 ortholog call to YPR116W robust beyond sequence similarity (synteny/structure)? (2) Should the inviable phenotype be reconciled with a respiratory-only role? (3) Is there any *S. pombe* Rpm2 ortholog for Rrg8 to associate with?
- **Suggested experiments:** the five Discriminating Tests above (complementation and mito pre-tRNA assay are highest value).

Provenance: All database values were fetched programmatically this run (UniProt REST for O14106/Q06109; PomBase API `annotation_details` for SPAC31G5.06; SGD backend for S000006320; EBI InterPro API). Sequence identity was computed by an in-run Needleman–Wunsch alignment (65/280 = 23.2%). Where a resource returned no content (InterPro for O14106, HTTP 204), this is reported rather than inferred.
